Year in Review—Heart Failure: Stability, Decompensation, and Readmission

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Chronic heart failure (HF) is a common diagnosis seen by physical therapists in many settings. This Year in Review article specifically focuses on current empirical evidence regarding escalating readmissions for HF and the potential role physical therapists can play in reducing these events. We examine the current HF readmission prediction models using the International Classifications of Function, Disability, and Health framework to highlight the need to consider reduced exercise capacity, ability to perform activities of daily living, and quality of life as integrated, inherent, and integral components of chronic HF itself. Consideration of stability at the levels of body structure and function, activity, and participation is proposed, and related literature is reviewed and integrated. **(Cardiopulm Phys Ther J. 2015;26:58–72)** *Key Words: heart failure, readmission, decompensation, stability*

INTRODUCTION

Heart failure (HF) is a major health concern. The prevalence of HF has dramatically increased in the United States and worldwide in the past decade. Three major factors have contributed to this increase in prevalence in the United States. These factors include aging of the population, a longer survival rate for patients who have sustained a myocardial infarction, and an increase in the prevalence of obesity and diabetes.[1]

Heart failure is estimated to affect approximately 5.7 million individuals in the United States with about 670,000 incident cases each year.[2] In addition, over 1 million hospital discharges were attributed to HF in 2007.[3] Heart failure is the most frequent cardiac diagnosis for hospital admissions and readmissions.[4] Therefore, the management of HF results in high levels of health care utilization and is a major contributor toward increasing health care costs. In light of rising costs and the impact of HF on reduced exercise capacity and quality of life, more effort is warranted to understand effective strategies for physical therapy contributions to the management of patients with HF.

The vision statement of the American Physical Therapy Association defines the need for therapists to transform society by optimizing movement to improve the human experience.[5] In this article, we review current empirical evidence published in 2014 on a national concern of escalating HF readmissions and the potential role physical therapists can play in reducing readmissions. Author commentary on selected articles published during 2014 highlights considerations regarding the physical therapist's role in the prevention of readmissions in patients with HF. Commentary will focus on the prevention of exacerbation by improving patient stability. We specifically identify areas of consideration related to this theme and selected research publications for each area. This is not a systematic review and so we make no claims that the selection of the set of articles is free from bias. We simply put forward thoughts for reasonable consideration and we welcome dialog.

The International Classification of Functioning, Disability, and Health (ICF) as part of the family of international classifications developed by the World Health Organization focus on human functioning.[6] The use of the

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ICF model provides standardized terminology and a framework that centers on the functional aspects of life. The consideration of the ICF model allows physical therapists to effectively witness and consider the many factors that contribute to a patient's treatment and recovery.

When considering HF in light of the ICF model, reduced exercise capacity, ability to perform activities of daily living (ADLs), and quality of life should not simply be considered disease outcomes as a consequence of HF. These 3 interrelated factors are inherent and integral parts of the disease itself and could be critical for considering strategies to reduce hospital admission and readmission in patients with HF. Exercise capacity, ability to perform ADLs, and quality of life should not be considered isolated factors related to readmissions. In this review, we consider how exercise capacity, ADLs, and quality of life interact to contribute to patient stability. Simply considered, a lack of physiological stability results in challenges to homeostasis. To maintain homeostasis with less physiological stability, a person must be able to adjust behavior. For example, for people native to high altitude, there are less behavioral adjustments needed during a high-altitude ascent than for a lowlander. A lowlander will need to adjust behaviors, move slower, take more rest breaks, and perhaps use oxygen. The native has greater physiological stability to meet the altitude-related challenges to homeostasis.

This article presents recent findings about the interacting concepts of decompensation, readmission, and stability in patients with HF. The underlying premise is that a reduction in stability increases the risk of decompensation, and decompensation leads to hospital admission. This article will first define stability, decompensation, and readmission. We then review selected articles that relate to these 3 concepts. Throughout the article, commentary is provided about how these concepts are integrated and what they suggest about strategies to reduce hospital readmission.

KEY CONCEPTS

Stability

We define stability to mean physiological stability as generally as possible. This includes what comes to mind when we consider stability as a feature of posture and balance. In addition, physiological stability includes stability of all the physiological processes involved with homeostasis. For example, consider postural stability as related to homeostasis. Without postural stability, there cannot be skilled movements, and without skilled movements, we cannot seek out shelter, food, and water, all of which are necessary physiological processes to maintain homeostasis.

The hierarchy of physiological systems contains several nested processes that enable higher levels of control and order. Stability of one level of organization enables a higher level of organization, which then becomes a necessary part of maintaining stability of the lower levels. For example, organization and stability at the level of the excitable cellular membranes allow muscle and nerve function. Muscle and nerve as higher level organizational functions enable movement, which allow eating and drinking, which are necessary for the stability of membrane potentials through fluid and electrolyte balance.

There is one aspect of stability as described above to consider for our purposes. Stability requires control. Control is enabled in nested physiological systems through a variety of mechanisms, but common to all control mechanisms is the consideration of effectiveness. The effectiveness of a control system partly depends on what Ashby[7] referred to as requisite variety. Requisite variety refers to the range of responses that a control system has when faced with a challenge. If heart rate (HR) is part of the control system for blood pressure (BP) and there are only 2 possible HRs (50 and 150 beats per minute), then HR would have limited control of BP. What makes HR an effective part of the BP control system is that there is a large variety of HR options in the control of BP. At rest, we take advantage of the ability to adjust R-R intervals within milliseconds giving rise to HR variability. For example, Sloan et al[8] review the importance of both variance in cardiac autonomic control and HR on BP stability. Requisite variety is not simply the range of values. An HR of 50 and 150 are quite far apart in the spectrum of reasonable HRs. But the spacing, the number of discrete options within the range, is important for requisite variety.

In patients with HF, there are challenges to physiological stability, and these challenges come in the form of reduced requisite variety. Patients with HF have a lower dynamic range of physiological control and possibly less discrete options within that range. As described above, nested physiological processes of lower levels enable higher levels of organization, which allow the maintenance of the lower levels of stability. With HF, there are problems with lower level stability that limit higher levels of organization, placing challenges on the higher level ability to maintain lower levels. This cyclic process results in ever decreasing requisite variety, an inability to maintain physiological stability, and eventually leads to decompensation. Figure 1 illustrates the emergence of lower to higher levels of physiological stability. Several "low-level" physiological variables are considered relating to oxygen consumption (V_{O_2}) and therefore work capacity. Work capacity and skills allow ADLs to be performed (therefore meeting basic, instrumental, and life needs, including leisure and rest). We realize that being able to do ADLs does not guarantee quality of life nor does being able to do ADLs mean there will be low quality of life. But "unmet ADLs" will, at the very least, make it more difficult to have life quality and will also make it more difficult to maintain lower levels of stability. For example, if nutrient intake is reduced, then hemoglobin levels can fall or if rest is not possible, then the range of HR can be compromised or skills can be reduced from mental fatigue.

This is why we propose that reduced exercise capacity, ability to perform ADLs, and quality of life not simply be considered consequences of HF or isolated causes of HF admission. Exercise capacity reflects higher level

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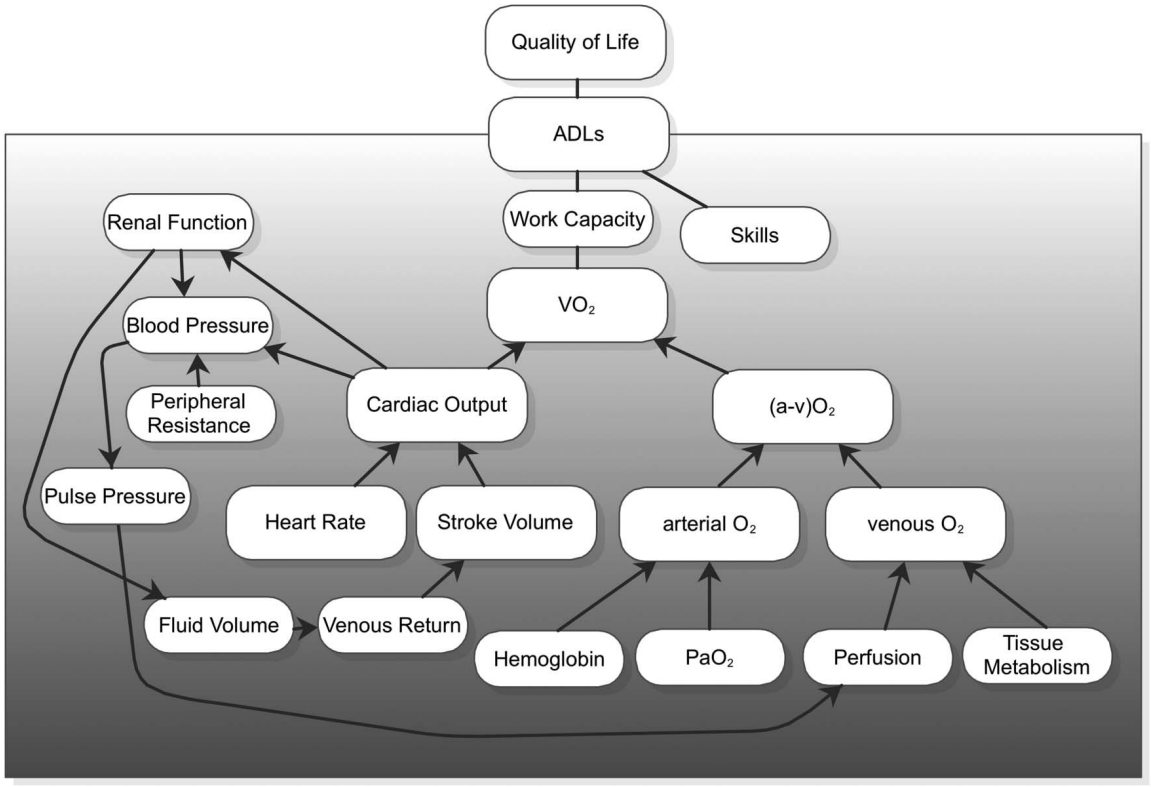


Fig. 1. A representation of levels of stability from basic physiological processes related to heart failure to Vo_2, work capacity, ADLs, and quality of life. Abbreviations: ADLs, activities of daily living; O_2, oxygen; PaO_2, partial pressure of arterial oxygen; Vo_2, oxygen consumption; $(a\text{-}v)o_2$ = arterial venous oxygen difference.

organization and the ability to do work. It is a necessary condition for performing ADLs. With a reduction in the requisite variety of cardiac output (lower range through a reduced maximum), there is a lower range of exercise capacity. If this is combined with additional changes in requisite variety (muscle strength, ventilation, etc), there is an even greater restriction of the requisite variety of exercise capacity. If this restriction of requisite variety of exercise capacity is severe enough, it can influence the ability to perform ADLs. Being able to do ADLs then represents an even higher level of organization requiring both the ability to do work and knowledge to perform the tasks. Occupational tasks, ADLs, are more complicated for someone with HF because of their reduced exercise capacity. To maintain stability, patients must compensate for these limitations with more behavioral requisite variety. However, this is challenging because the behavioral requisite variety requires exercise capacity. Therefore, to increase behavioral requisite variety, patients must adapt and behave differently. They need to learn new ways of performing ADLs because ADL needs being met are a necessary condition for quality of life. Quality of life represents an even higher level of organization that includes the concepts of self-efficacy, hope, will, and desire. For example, we would propose that without these concepts, it is much more difficult to attempt to learn new approaches to ADLs (self-care that includes adaptive skills, resources, and environments), and without some ADLs, it is difficult to maintain exercise capacity. If the ADL that is no longer effective is related to nutrition, then it becomes ever more challenging to maintain the lower levels of stability. As lower level systems become unstable (ie, cardiac output is reduced to the renal system), then the requisite variety for cardiac output is further restricted by increased afterload. This further reduces exercise capacity and ADLs. To break the cycle, a new approach to behavior, in part enabled by quality of life and resources, is required. However, as this review will demonstrate, the current mode of prescribing medical fixes tends to lead to incrementally shorter periods of time of stability in patients with HF.[9] Figure 2A is a system with requisite variety of Vo_2, Work Capacity, Skills/Resources/Environment, ADLs, and Quality of Life. In Figure 2B, we show a system with reduced requisite variety and the same Skills/Resources/Environment. The limited Vo_2 then limits the system all the way up. In Figure 2C, we see an adaptive response when Vo_2 and Work Capacity cannot be altered. For more requisite variety at higher levels, which then assist in the maintenance of lower level stability, the system needs additional requisite variety in Skills/Resources/Environment such as new methods, new skills, an adaptive system, and environment.

Decompensation: Exacerbation

We will use the definition put forth by Felker et al[10] for decompensated HF: "new or worsening symptoms/signs of dyspnea, fatigue, or edema, leading to hospitalization or unscheduled medical care (such as unscheduled visits to clinics or emergency departments)." We consider the

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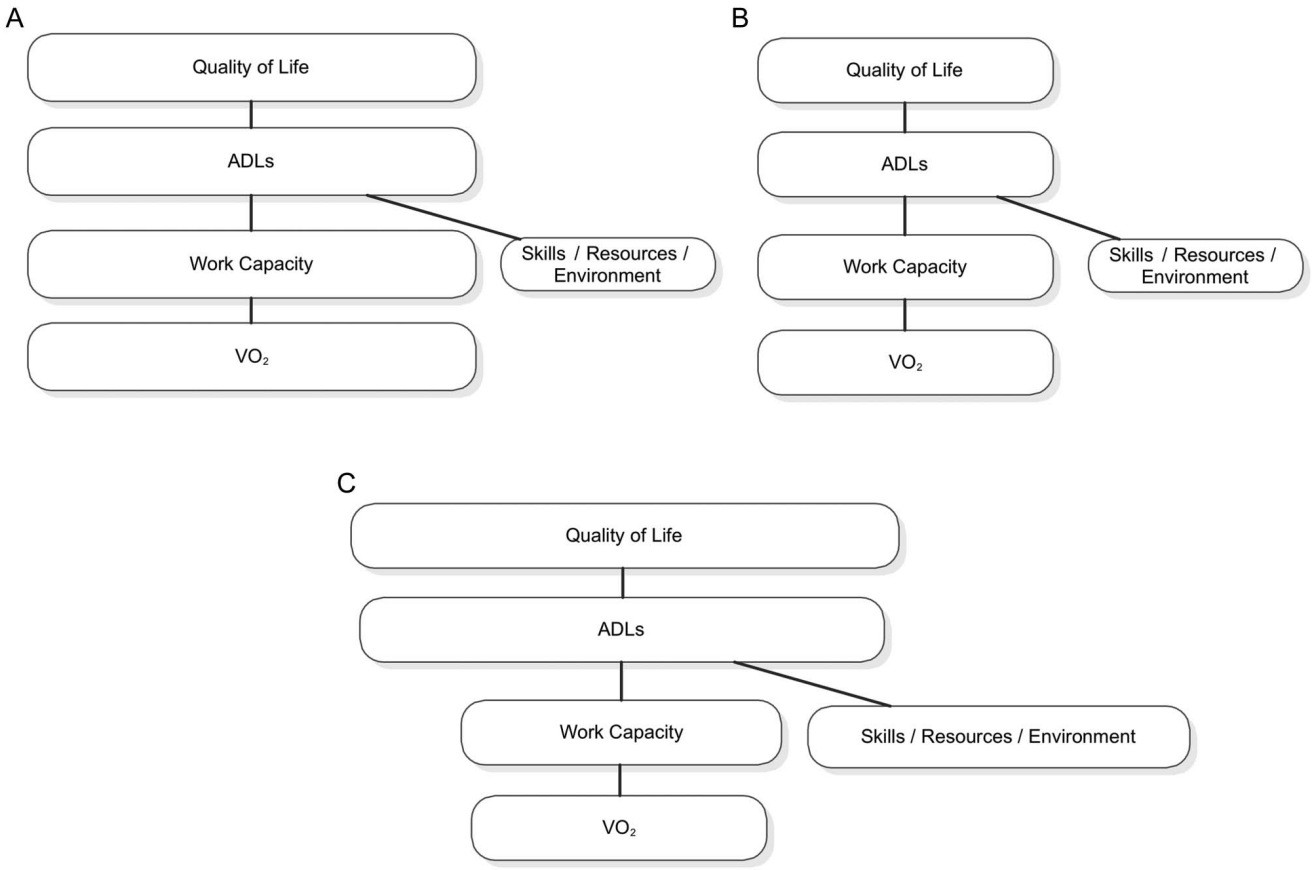


Fig. 2. A representation of requisite variety at various levels of stability. In (A), there is a broad range of variety (width of the boxes). From Vo_2 upward, there is a variety for response. In (B), there is a reduction in response, which necessarily reduces response variety at levels above; unless, in (C), there is an infusion of new requisite variety including skills, resources, and environmental factors. Abbreviations: ADLs, activities of daily living; Vo_2, oxygen consumption.

antecedent: "new or worsening symptoms/signs of dyspnea, fatigue or edema" as a starting point for defining what is happening in a patient with chronic HF becoming decompensated from what has been otherwise compensated. Compensated is simply "not decompensated" that includes "relatively" stable symptoms/signs (not new or worsening). When patients with chronic HF have "new or worsening symptoms/signs," they are heading toward decompensation. This occurrence of new symptoms/signs (decompensation) is often a reason for unscheduled visits, including readmission to the hospital. Therefore, the consequence (leading to hospitalization or unscheduled medical care) is considered part of the definition of decompensated HF. When someone with chronic HF is decompensated, the more precise terminology proposed is that the patient has an exacerbation.[10] This classification simply delineates this group of patients from those with acute (new-onset) HF.

People with a history of chronic HF can have new symptoms/signs that do not provoke an unscheduled visit, and although these should not be ignored, such new symptoms/signs are not considered exacerbation because there was no unscheduled visit. This approach to defining decompensation/exacerbation provides boundaries on relative stability in a person with chronic HF. Symptoms/signs may vary within a range of relative stability, but if they vary to the point of an unscheduled visit, then the patient is experiencing an exacerbation. Because of the high prevalence of HF relative to its incidence, exacerbation is the primary cause of HF admissions and all HF-specific readmissions.

Readmission

Readmission is simply being admitted to the hospital within some period of time of an index (first, incident) admission. It is important to keep 2 factors in mind: the period of time and the readmit diagnosis. It is easy to imagine how the readmit period of time has a major influence on reported rates and predictors. For example, the lifetime readmission rate for HF likely approaches 100% because only patients who suddenly die out of the hospital would never have a readmission if considering their entire lifetime. The time frame of readmission influences what might be considered reasonable explanations and predictors of such readmissions.

The second factor is readmit diagnosis. All-cause (crude) readmission rates are admissions during the allocated period of time with any diagnosis. Medicare (Centers for Medicare & Medicaid Services, CMS) definition, reported readmit rates, and readmit prediction models are based on all-cause readmissions.[11] If a patient is admitted with HF, is discharged, and is then readmitted after a fall 1 week later,

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then it is a crude readmission. If readmitted due to HF exacerbation 1 week later, then it is a disease-specific readmission (in this case, an HF-specific readmission). These are important factors to keep in mind when considering the published literature on readmissions generally and also when considering preventive measures. For example, if the all-cause readmit rate for HF is much higher than the HF-specific readmit rate then preventive measures should be focused on prevention of the highest occurring readmit diagnosis aside from HF. However, if the HF-specific readmit rate is close to the all-cause readmit rate, that is, HF diagnoses account for a large proportion of readmits, then prevention efforts should be aimed at reducing the risk of HF readmission. It is quite possible that there are regional and institutional variations in the readmit rates based on variation in patient care and patient social factors.

REVIEW OF LITERATURE

As indicated in the introduction, this is not a systematic review. However, we do believe it is helpful to the reader to know the process that was undertaken. There were 3 approaches to gathering articles for inclusion. First, the table of contents of several cardiology and cardiac rehabilitation journals were reviewed for the year 2014. Article titles that indicated the article may provide insight in the theme of this review were collected for consideration. Second, a literature search on "heart failure and X" with X being decompensation, stability, or readmission was performed in Google Scholar and limited to "since 2014." The search was repeated until early 2015. Third, the personal libraries of both authors were reviewed. Because both authors are actively involved in projects related to the topic of this review, the 2 libraries provided approximately half of the included articles being reviewed. Articles were selected based on the opinion of the authors that the article was relevant to the theme (first necessary criteria) and of sufficient quality (second necessary criteria) to be included in the review. While we came across several very interesting abstracts from symposia and conferences, we have decided not to include these. We hope to see many of them in upcoming years for future reviews on this topic.

The following review starts with readmission although we believe the causal pathway is from a lack of stability to decompensation to admission/readmission. This reverse order is followed because we believe that characteristics of patients being admitted and readmitted provides insight into the relevant features of decompensation to be considered, and features of decompensation provides insight into the concept of stability.

READMISSIONS

Given the timeliness of the topic of HF in general and readmissions in particular, several models have been published aiming to predict HF outcomes, including the risk of hospital admission or readmission after an index admission. Models that can discriminate between patients at risk for hospital admissions or readmissions are valuable for decision making regarding resource allocation for prevention. It is also possible that such models would lead to insights regarding the underlying causal mechanisms relating HF to hospitalization. Classifying patients based on the underlying causal mechanisms would allow for the development of targeted prevention interventions. Such models could also help with the design of future studies with patients. Studies could target subjects with a higher risk of admission to increase the likelihood of events during a trial period. There have been so many models that 2014 witnessed the publication of systematic reviews of readmission prediction models. In this section, we review 2 such systematic reviews,[12,13] 3 recent prediction models each adding a unique characteristic for consideration,[9,14,15] and a systematic review of general interventions to reduce readmissions in patients with HF.[16]

Systematic Reviews of Prediction Models

Systematic reviews of prediction models offer insights into meta-trends relating model characteristics and predictive discrimination. A model must first be predictive within its development sample and provide a reasonable degree of predictive discrimination. A common statistic to evaluate and compare predictive discrimination of models is the C-statistic. Discrimination is how well a statistical model can distinguish subjects that experience an outcome. A C-statistic of 1 indicates perfect discrimination, whereas a C-statistic of 0.5 indicates discrimination no better than chance.[17] If a model has sufficient predictive discrimination in its development sample, it should then be validated with a sample separate from its development.

Ouwerkerk et al[12] reviewed outcome prediction models to report on the strongest variables and model characteristics in patients with HF. The authors included models with mortality, hospitalization, and mortality or hospitalization as predictor outcomes in both chronic HF and acute decompensated HF patient populations. They identified 117 models in 55 articles that included 249 different variables. The overall range of C-statistics for reported models was 0.52 to 0.90.[12] The highest of these values was a model predicting in-hospital mortality, whereas the lowest was predicting 6-month hospitalization. This is related to the overall trend whereby predicting mortality allows for more discrimination than predicting hospitalization and predicting shorter-term events allows for more discrimination than longer-term events. The overall mean C-statistic for predicting hospitalization was 0.71.

An important consideration here is that this article does not address whether the prediction models have been validated. It is one thing to develop a predictive model based on a set of data where the outcomes are known. Whether that model is predictive of outcomes in a different set of patients is very different but is a necessary part of model validation. The authors conclude that prediction models need to be improved before they will be helpful to clinicians and patients.[12]

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It is interesting to note that the only functional variable included in the 117 models reviewed was NYHA functional class. This system does not account for variations in living environment or necessarily relate unmet ADL needs, particularly completing ADLs with assistance, adaptive systems, or compensatory approaches. Future models for hospitalization (and readmission) should account for unmet ADL needs[18] and the postdischarge environment.[19]

Rahimi et al[13] also conducted a systematic review of published prediction models for HF outcomes to identify consistently reported independent predictors of risk.[13] Of the 64 models identified, 10 models were specifically related to hospitalization. Of these 10, 8 predicted all-cause hospitalization and 2 predicted HF readmission. As reported by Ouwerkerk et al,[12] models predicting death had a higher discriminatory ability than those predicting hospitalization.

Discriminative ability of the hospitalization prediction models varied with a C-statistic from 0.60 to 0.82. Recurrent predictors included age, sex, renal function, cardiovascular disease, and HR. As would be expected, the authors found that when different follow-up time periods were reported for the same models, the size of the C-statistic was lower for longer time periods.[13] In other words, it is more difficult to predict distant readmissions. Rahimi et al[13] conclude that there currently are no hospitalization risk models of sufficient performance for wide clinical use.

There are some useful trends identified from these 2 systematic reviews of models attempting to predict hospitalization. First and foremost, it is difficult to predict hospitalization even in large samples with known outcomes from large sets of predictive variables. Second, models predicting longer follow-up time periods have lower discrimination. We also notice that models currently available do not represent several relevant social environmental factors related to ADL needs (or unmet needs).

Published Prediction Models

Three new prediction models specifically for readmissions in patients with HF were published in 2014. Each article offers a perspective slightly different from previous models. The CMS has an HF readmission prediction model (www.readmissionscore.org) as described by Keenan et al.[11] Its predictors are available from CMS claims. Based on the systematic reviews, claims data generally result in lower predictive discrimination than medical record data. The CMS readmission model has a C-statistic of approximately 0.58.[14] A benefit of the model, despite its low predictive discrimination, is that it is based on a large nationwide sample (~80,000 cases) from 4777 hospitals. Once developed, the model was validated with a nondevelopment sample of patients, which demonstrated similar predictive discrimination. Although low, the predictive discrimination is stable, likely due to the size of the development sample. The 3 models described below offer new approaches. Hummel et al[14] simply add hospital admissions before the index admission. Bakal et al[9] modeled the time between admissions across a 5-year follow-up period and identified predictors of reductions in the time before the next admission. Finally, Bayati et al[15] take a "big data" approach by considering more than 3000 predictors as part of a prediction-to-action data pipeline. The primary predictor variables identified in these models are presented in Table 1 for a broad overview comparison. For details on the models and statistical features of the predictors, the reader is referred to the primary sources.[9,14,15]

Those That Get Admitted a Lot Are More Likely to Get Readmitted. Hummel et al[14] took a clear and simple step to add to the predictive discrimination of the CMS model. The working hypothesis was that an admission within the year before an index admission would be predictive of a 30-day readmission. Using the Michigan Medicare Quality Improvement Organization, they analyzed Medicare patients aged 66 years or older (to ensure 12 or more months of claims data for prior admissions). The incremental effect of adding prior admissions to the CMS model was tested. During the study period, there were 1807 index hospitalizations. Of these, 1018 (56.3%) had not been admitted in the previous 12 months, 254 (14.1%) had been admitted once, and 535 (29.6%) had been admitted 2 more times. Of the entire sample, ejection fraction (EF) data were available in 80% of the subjects (1438).

Applying the CMS model to the Michigan cohort resulted in a similar C-statistic (0.61) to the original CMS model report, with higher predictive discrimination in patients with preserved EF than those with reduced EF (0.67 vs 0.60). The predictive discrimination of prior admissions alone was higher than the CMS model in this sample (0.71 vs 0.61). Prior admissions combined with the CMS model resulted in a C-statistic of 0.74 for a 30-day readmission. The odds ratio (95% confidence interval, CI) for the prior admit and CMS model for patients with 1 prior admit was 4.67 (3.37–6.46) and 2 or more prior admits was 6.49 (4.93–8.55).

The authors believe that history of admissions is a proxy for complex socioeconomic factors that may vary between hospitals.[14] They have not validated the model in a nondevelopment sample. They have not tested whether patient or family recall as a source of prior admission status is consistent with Medicare claims confirmation (which may not always be available). This simple change in the CMS model had a significant impact on predictive discrimination. It is possible, as the authors point out, that prior admissions capture factors such as social environmental issues related to ADL needs. It is harder to imagine, but possible, that these factors are not already being addressed in patients with a history of frequent admissions. It is also hard to not simply pass off these findings as a statement of the obvious, patients who get admitted more often are more likely to get readmitted. To break out of this circular reasoning, we now consider a different model.

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TABLE 1

Primary Predictors of 30-Day Heart Failure Readmission From Prediction Models

	Bakal et al[9]	Bayati et al[15]	Hummel et al[14]
Predictors of increased risk	Charlson score	Low lymphocytes	Model with prior admits
	ED visits 180 days before	**Admitted in past 6 months**	**Prior admissions**
	Age	High BUN	Low sodium
	Length of stay (per day)	Low monocyte	Cardiac arrest
	See note	History of nondependent abuse of drugs	Model without prior admits
		COPD	History of HF
		Gastrointestinal hemorrhage	Low sodium
		Elevated AST	Low HcT
		Cardiomyopathy	
		Low magnesium	
		Elevated INR	
		Isolated hospital room	
		High BNP	
Predictors of decreased risk	Male	**Number of ED visits during past 6 months <2**	
		Normal HcT	
		Normal BNP	
		Normal alkaline phosphatase	
		Normal chloride	
		Not on digoxin	
		Low MCHC%	
		Low TSH variation	
		Low CO_2 variation	
		Low RDW variation	
		Low MCV variation	

For Bakal et al,[9] variables are from the gap-time model and predict a reduction in the time (or an increase in time) between repeat hospitalizations; for example, elevated Charlson score decreases the time between admissions, while being male, increases the time between repeat admissions. For Hummel et al,[14] only variables with odds ratios (95% CIs) that do not include the null value of 1.0 are included. For all models, variables are listed in descending order from the strongest predictors to the weakest; variables common to all 3 models are in bold.

Abbreviations: AST, aspartate aminotransferase; BNP, brain natriuretic peptide; BUN, blood urea nitrogen; COPD, chronic obstructive pulmonary disease; CO_2, carbon dioxide; ED, emergency department; HcT, hematocrit; HF, heart failure; INR, international normalized ratio; MCHC%, mean corpuscular hemoglobin concentration; MCV, mean corpuscular volume; RDW, red blood cell distribution width; TSH, thyroid-stimulating hormone.

Once People Start Getting Admitted, Additional Admissions Happen Sooner. Bakal et al[9] add a very interesting component to the process of predictive modeling for hospitalization, one that resonates quite well with common sense and clinical intuition. This study compared repeat hospitalization models in a population-based cohort of 40,667 patients for both HF and all-cause hospitalizations with up to 5 years of follow-up. The basic idea is related to that of Hummel et al[14] that sick patients are readmitted more frequently. They compared a gap-time model to estimate the adjusted time between hospitalizations.

Models were adjusted for age, sex, length of stay, acuity of admission, Charlson index (providing a weighted score based on the number and type of comorbidities), and number of ED visits in the previous 6 months. The gap-time model demonstrated that during the follow-up period of 5 years, about half of the sample would return for a hospitalization after each repeat hospitalization. The time between hospitalizations decreased by 40% with each repeat admission. Furthermore, each admission was associated with a 28-day (95% CI, 22–35) reduction in time out of the hospital. Patients who were hospitalized more often for HF tended to be older and were more likely to have comorbidities such as diabetes, peripheral arterial disease, and chronic obstructive pulmonary disease but not dementia or cancer. This study demonstrates that the time between successive admissions decreases with each hospitalization. This pattern is present even after adjusting for prognostic factors such as age, number of ED visits, and the Charlson index.[9]

The approach taken in this article is novel in application and potentially fruitful. The authors recognize that additional data would make predictions more reliable such as including clinical (physiological) data. Like the model described by Hummel et al,[14] the model by Bakal et al[9] uses information about previous admissions to forecast future

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admissions and is consistent with clinical intuition. However, unlike the Hummel et al[14] model, the Bakal et al[9] model is more dynamic and offers insights about patterns of stability in patients with HF. Interventions should aim to extend the time frame between admissions and effective interventions to reduce admissions based on history. For example, if patients have only been out of the hospital for 60 days on the index admission where an intervention is being tested, then keeping more than half of them from being admitted for another 60 days would be an improvement based on the predictions of the Bakal et al[9] model.

Big Data Pipeline for Prediction and Action. Bayati et al[15] constructed a predictive model for readmissions in patients with HF and studied how predictions could be used to guide patient-specific interventions. They referred to the system as a prediction and action pipeline. A particularly unique feature of the model development was an assessment of the cost-effectiveness of combining prediction and decision making to determine interventions. The authors point to the prohibitive expense of applying intensive postdischarge interventions to all patients with HF. Using deidentified patient data drawn from the electronic health records system for a tertiary urban hospital serving the Washington, DC, area, they developed a model from a retrospective database of 793 hospital visits for HF to predict whether patients would be readmitted within 30 days of discharge. They also performed a cost-effectiveness analysis of 379 additional hospital visits that were not included in either the development sample or the decision analysis.[15]

Variables shown to increase or decrease risk of readmission are presented in Table 1. The big data approach does have a benefit of identifying potentially new insights. These new insights require careful consideration because the models themselves do not allow causal inferences to be drawn and cannot be used to plan new interventions. As an example, a predictor identified in this model was "Patient has been in isolated room in hospital."[10] This is most likely a proxy for having an infectious process. Therefore, an intervention that should not be considered to reduce readmissions is avoiding isolation rooms in patients with HF. Overall, the model developed by Bayati et al[15] found that readmissions could be reduced by approximately 18% and the associated cost could be reduced by approximately 4%. They conclude that these approaches are particularly useful when it is not economically feasible to provide interventions to all patients. Of course, the model proposed does not provide a suggested intervention based on predictors, aside from current practice of attempting to stabilize indicators of disease severity, particularly one that could be shown to reduce the rate of readmissions by 18%.

General Interventions to Reduce Readmission Risk

Feltner et al[16] completed a systematic review of 47 randomized controlled trials of general interventions to reduce readmissions in patients with HF. These interventions were not necessarily tailored to specific patient needs nor were they applied to selected high-risk groups (other than the baseline high risk of having chronic HF). The clearest outcome was that interventions based on home visits were superior to other forms (telephone, telehealth, and clinic) and that more intensive interventions (more frequent) were more effective than less intensive interventions. There was a pooled reduction of risk for 3- to 6-month all-cause readmission for home care interventions after hospital admission (relative risk [95% CI] = 0.77 [0.66–0.86]).[16] Home care was the most effective intervention, followed by a clinic-based follow-up. They reported on one home care study that reduced risk for the 30-day readmission period. This was considered a "high-intensity" intervention as it consisted of 8 planned visits within the first 24 hours after discharge. Telehealth, structured phone interview, cognitive, or educational-only interventions did not have a statistically significant reduction of readmission risk for either 30 days or 3 to 6 months.[16] It is interesting to consider these results in light of the Bakal et al[9] gap-time model demonstrating that patients with HF have ever shortening periods of time between subsequent hospitalizations. As previously discussed, this suggests that there are real changes in the patients' stability over time making staying out of the hospital increasingly difficult. The challenges for an intervention to demonstrate a reduction in 30-day readmission risk may be due to this fact. However, with a high enough intensity intervention, this risk can be reduced. The question is whether a prediction model could predict those patients likely to return within 30 days accurately enough to allow such a high-intensity intervention to be cost-effective.

The findings of this systematic review are consistent with a more recent randomized controlled trial assessing the effectiveness of educational interventions regarding self-care in patients with chronic HF in rural areas.[14] In 602 patients, there was no significant change in the combined end point of cardiac death or HF hospitalization for patients receiving a self-care educational intervention during a 2-year follow-up period. Therefore, increasing the frequency of patient–clinician contact for education sessions does not seem to reduce the risk of readmission in patients with HF.

Summary of Readmission Prediction

This past year witnessed significant activity attempting to increase our understanding of readmissions in patients with HF. The benefit of better prediction would be resource allocation and targeted interventions to prevent readmissions. The overarching message at this point is that prediction is not yet discriminatory enough for wide-scale clinical use, but there are trends emerging that warrant consideration. The finding of Bakal et al[9] is particularly insightful, and we believe it will resonate with physical therapists' clinical observations. Patients with chronic HF tend to get admitted to the hospital with reduced gaps

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between admissions and hence an increasing frequency over time. This finding helps explain the presence of emergency department (ED) visits or prior admissions in the other new models. In fact, as seen in Table 1, prior admissions (or ED visits) is the single common predictor variable across the 3 new models from 2014. A major benefit of the gap-time model by Bakal et al[9] is that it helps us step away from the seeming circularity of this fact (that frequent admissions are predictive of frequent admissions). It is not simply that frequent admissions are predictive admissions but that with each admission, something is different that then tends to shorten the time before the next admission. We cannot claim to know the causal mechanisms here, whether each admission causes a change that increase risk or whether something changes in the person that shortens the gap between readmissions. However, we can consider explanations.

The following possible explanations are not mutually exclusive. One explanation is that over time, the progressive nature of HF leads to the patient getting worse, having a lower physiological stability, and less requisite variety at a primary level purely due to disease progression. With less stability, there is less room for error in the medical and behavioral management resulting in a narrower "safe zone" before triggering a cascade of decompensation and new symptoms/signs that result in an unscheduled visit. If this is the case, then prior admissions and/or ED visits are acting as a proxy for disease severity and potentially reduce the predictive power of these other variables when taken singularly. This is similar to colinearity in a multiple regression model. When predictor variables that are related to one another are simultaneously considered, then one diminishes the predictive ability of the second. The information from the second is redundant with that of the first (ie, contributes no new understanding). It may be that the power of prior admissions' predictive ability is redundant with many other indicators of worsening disease status but because it is focused information, it prevails as the stronger predictor. Ironically, other variables—taken singularly or at least chosen for a clear physiological reason—would be needed to develop targeted medical interventions. Hummel et al[14] seems to refute this explanation by presenting the CMS model fully with and without prior admissions and demonstrating prior admission as the strongest predictor. But the power of prior admission may be its inclusion of several factors that far outweigh any single predictor even if redundant.

Another possible explanation is that, as Hummel et al[14] propose, prior admissions account for unmeasured socioeconomic (or, as we proposed, social environmental) factors not otherwise accounted for in the models. This is the new information perspective that adding prior admission to prediction adds new information leading to stronger models. Unmet ADL needs are not considered in any of the models but has been demonstrated to be a predictor of admission risk in the elderly.[18] If unmet ADL needs increase risk of readmission in patients with HF, because it is not accounted for by other variables, then its effect would be accounted for by frequency of prior admissions. In this case, prior admission may act, in part, as a proxy for unmet ADL needs. However, if part of the power of prior admissions in a prediction model is that the concept includes unmet ADL needs, then keeping this in the prediction model as prior admissions, and not as unmet ADL needs, eliminates the potential to identify unmet ADL needs as a specific cause or readmission. The benefit of identifying a specific cause is that interventions can then be targeted toward that specific cause. It is not difficult to imagine that a targeted intervention directed at a specific causal mechanism has a greater chance of improving outcomes. What is clear, as previously stated, is that simply increasing the educational points of contact as nontargeted interventions in patient care does not reduce readmissions.

In summary, although prior admissions are strong predictors for readmission and the gap-time pattern of HF admissions demonstrates shorter periods before the next admission, it would be far more valuable for targeting interventions if models were developed in an attempt to understand the mechanisms underlying the repeat admissions. Targeted interventions may involve identifying which patients need frequent visits and what those visits should include for each specific patient (or at least classes of patients). As shown in Figure 3, there is a need to consider the cycle of admissions as a temporal trend reflecting the shortening of time between each cycle of stability, decompensation, and admission. Also, rather than keeping prior admissions as a proxy for socioeconomic and/or environmental factors such as unmet ADL needs, it would be more helpful to develop models that included such variables. If the only thing we are targeting is a history of prior admissions in an attempt to reduce readmissions, then we may find ourselves chasing our tail.

STABILITY

We now consider the literature from 2014 that provides insights related to stability and factors influencing stability in patients with HF. When we say a patient with HF is stable and can maintain stability, we are simply stating that they are compensated and can maintain the compensated state. Being compensated and maintaining the compensated state are different considerations. A patient may be compensated but not able to maintain a compensated state given a certain factor. For example, a patient may be stable but not able to maintain the compensated state given a dietary increase in sodium of more than 1500 mg. Whether a patient rebounds from an event (ie, too much sodium) that challenges stability has to do with both physiology (how much requisite variety of effective response) and behavioral adjustments (again, requisite variety of effective response).

Levels of Stability

We will consider levels of stability based on the ICF model. The first level is directly related to ICF body structure

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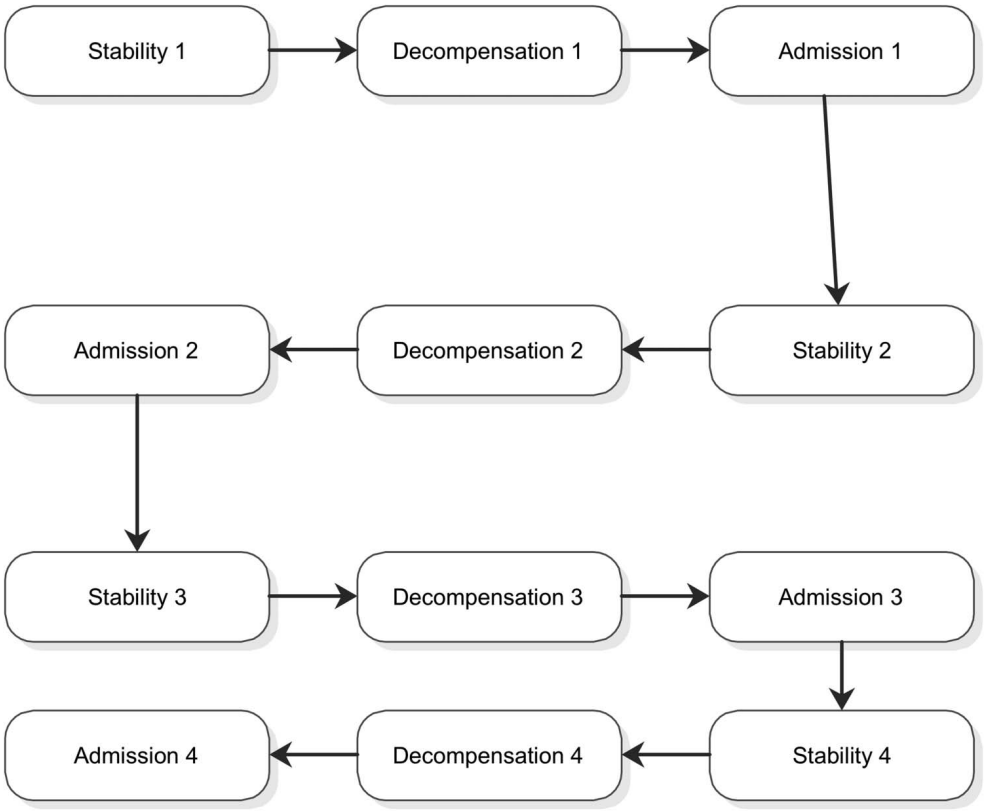


Fig. 3. A temporal sequence of new and decreased levels of stability that when reduced leads to decompensation and hospital admission. Between each admission, there is less time before the next.

and function. This level includes whether the patient is currently physiologically stable/compensated and includes having the requisite variety to maintain stability under varying conditions. The body structure and function level includes exercise or functional work capacity; hence, whether the patient's dynamic range of physiological response is large enough to accommodate fluctuations in challenges. The activity level of stability is whether the patient has the requisite variety to maintain body structure and function. As indicated in the ICF model, body structures and functions enable activities and activities support body structures and functions. This includes whether the patient can perform the required ADLs to maintain functional work capacity. Finally, the participation level includes work capacity and activities. This level considers whether the patient has the requisite variety in participation to allow activities. This requires the ability and desire to learn new things. We see this as being related to cognitive function, quality of life, self-efficacy, hope, and desire.

Body Structure and Function. The first level is whether the patient is physiologically stable, that is, currently compensated. Patients with chronic HF are in varying degrees of compensation. The hallmark signs of decompensation in a patient with chronic HF are those related to increasing congestion, in particular elevated ventricular cardiac filling pressures, and as aptly stated in a recent editorial, in this regard, "Observation is Never Obsolete."[20] Therapists should develop, practice, and refine their abilities of physical examination related to the classic signs of congestion: jugular venous distention, pulmonary crackles, and the S3 heart sound. These signs are most directly indicative of elevated cardiac filling pressures when compared with other signs of congestion such as peripheral edema and variations in body weight. Pulmonary crackles and an S3 heart sound are present in more than half of the patients presenting to the ED for HF decompensation and have prognostic significance.[21] Patients with pulmonary crackles and an S3 heart sound had higher readmission rates than those without (odds ratio for pulmonary crackles = 2.8; S3 = 2.6).[21] In another study, patients with orthopnea had a 12.8 (1.7–99.7) times risk of readmission within 6 months than patients without this symptom. This was a small study (n = 32), which is why the 95% CI is so wide. In the same sample, each decline in NYHA functional class was associated with a 3.4 (1.4–8.5) risk, making the presence of orthopnea a stronger predictor than a change from NYHA functional classification of II to IV.[22] It is important to consider these values as dynamic as compensation and decompensation are dynamic states. Therapists need to consider these values with their patients at each visit to notice changes. Compensation is stability, and in chronic HF, stability is relatively stable congestion. By relatively stable, we mean that if today they do not have crackles and remain relatively stable, then they should not have crackles by the next day also.

This past year witnessed the dawn of a new symptom of HF. Thibodeau et al[23] examined the frequency and hemodynamic correlates of shortness of breath when

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bending forward, which they named "bendopnea." In a prospective study of 102 subjects with systolic HF referred for right heart catheterization, the time to onset of bendopnea was measured. Bendopnea occurred in 28% of subjects with median (25th, 75th percentiles) time to onset of 8 (7, 11) seconds. Patients with bendopnea had elevated supine right atrial pressure and pulmonary capillary wedge pressure when compared with those without bendopnea, and they were more likely to have other symptoms of decompensated HF such as angina, dyspnea on exertion, orthopnea, paroxysmal nocturnal dyspnea, and abdominal fullness. The one sign that patients with bendopnea had that those without bendopnea did not show was JVD. An important factor is that patients with bendopnea had a higher BMI but lacked difference in waist/hip ratio or waist and hip circumference. Thibodeau et al[23] believe that awareness of bendopnea should improve noninvasive assessment of hemodynamics in subjects with HF.

Whether a patient has the requisite variety to maintain stability under varying conditions can be considered based on the presence of signs and/or symptoms during activities such as exercise or ADLs. This is the classic "stress test" approach to evaluating cardiac function. Can the heart keep up with increasing levels of demand? In other words, does the heart have the requisite variability to meet the needs associated with the exercise? Physical therapists consider whether patients have the physiological capacity to stay compensated under different situations. A physical challenge serves as a test, and the changes in signs and symptoms with such a test are an indication of whether a patient can remain compensated. Perez-Moreno et al[24] studied the prognostic value of the symptoms of fatigue and dyspnea. They measured fatigue and dyspnea relative to the workload that elicited the symptoms (much like the NYHA functional class considers symptoms). Eliciting these symptoms with gradations of lower levels of workload was a clear prognostic indicator. In particular, a change in fatigue symptoms with the same workload in a 6-month period was associated with a 55% increase in risk for a hospital stay (hazard ratio [95% CI] = 1.55 [1.24–1.94]).[24] A change in dyspnea symptoms with the same workload in a 6-month period was associated with an 88% increase risk for a hospital stay (1.85 [1.50–2.37]).[24] This study highlights the importance of recognizing changes in symptoms over time to recognize risk and with further work to validate tools with sufficient specificity and sensitivity will be an important component to physical therapist contributions to the overall care of patients with HF.

What is not known from the Perez-Moreno et al[24] study is whether the dyspnea and fatigue changes over time were associated with underlying irreversible changes in the cardiac pathology. There was an association whereby patients with higher symptoms had elevated NT-proBNP. However, NT-proBNP is simply an indication of elevated filling pressures due to congestion. It is possible that reversible physiological changes reduce requisite variety of physiological response that then lead to decompensation with subsequent elevations in NT-proBNP and symptoms. In other words, the presence of this laboratory sign along with the symptoms simply relates the symptoms to decompensation and not to the irreversible progression of cardiac pathology.

Seo et al[25] demonstrated a relationship between leg extension strength as measured by a handheld dynamometer and dyspnea. Subjects with higher leg extensor strength (by median split) had lower levels of dyspnea with ADLs, instrumental ADLs, and physical function. The stronger subjects also walked on average 49.43 m further on the 6-minute walk test. Patients with higher leg extension strength were more likely to be in NYHA functional class II, as opposed to class III or IV.[25] This was a cross-sectional study, and it is impossible to know the causal mechanisms at play. It is plausible that higher leg extension strength results in less symptoms and therefore might actually contribute to the NYHA functional classification; but it is also plausible that worse cardiac function associated with a higher NYHA functional class and more symptoms restricts movement and leading to less leg extension strength. Clinical trials that classify patients based on their baseline symptoms and leg extension strength and then perform muscle strengthening to test the magnitude of effect in those weak vs. stronger at baseline would be helpful to tease apart these causal pathways. It is also true that these pathways are not mutually exclusive. They could both be true, either in different patients or to a varying extent. For example, in Seo et al,[25] the 6 patients with stronger leg extension in NYHA functional classification IV are exceptions to the above causal pathways. These patients (although few) are stronger but continue to have more symptoms. It is more likely that their symptoms during activity are related to their cardiac condition rather than due to muscle weakness.

How much work is enough to increase and then maintain physical work capacity in patients with chronic HF? Ismail et al[26] recently reported on exercise training program characteristics and the magnitude of change in functional capacity in HF patients. Consistent with expectations, higher intensity resulted in greater improvements in physical capacity. Peak V_{O_2} for high-intensity training increased a mean of 3.33 (0.53–6.13) $mL \cdot kg^{-1} \cdot min^{-1}$, whereas for moderate intensity training, it increased 2.19 (1.39–2.99) $mL \cdot kg^{-1} \cdot min^{-1}$, and there was no statistically significant improvement in the pooled estimate from studies with low-intensity training.[26] Ismail et al[26] also demonstrated a moderate difference between studies requiring $>$460 kcal/week of load (2.58 [1.88–3.28] $mL \cdot kg^{-1} \cdot min^{-1}$) versus those with $<$460 kcal/week of load (1.81 [1.34–2.27] $mL \cdot kg^{-1} \cdot min^{-1}$). While high-intensity and/or high-volume training is relative to baseline, the subjects in all the included studies were clinically stable. The need for research to address exercise recommendations in patients with chronic HF that is relatively decompensated or recently have been decompensated has been made clear.[26] This past year, an abstract was presented on a clinical trial that tested the

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feasibility, safety, and effectiveness of a rehabilitation intervention in people with chronic HF who were in the acute care hospital for acute decompensation and during the immediate discharge period.[27] Preliminary reports of the results have demonstrated a significant improvement in the short physical performance battery, the 6-minute walk, and reductions in hospitalizations with no reported adverse events. We look forward for the full publication of results from this study for a future year in review and additional studies of patients with decompensated or recently compensated HF.

Activities. The ability to perform ADLs reflects the universal human condition as written about for centuries.[28] We must work to sustain ourselves. We work to live, and at times, it may feel as though we live to work. These ADLs are work tasks that fulfill a need. Physically healthy adults typically do not have a problem doing ADLs and fulfilling their needs. In fact, they can meet their basic and instrumental needs in a variety of ways. With a physical disease such as chronic HF, there are limitations placed on the work tasks themselves (what people are physically capable to do) and on the needs to be met (need to take medications, need to keep sodium intake regulated, need to keep fluid intake regulated, cannot tolerate altitude, cannot tolerate heat or cold, etc). Essentially the reduced requisite variety of body functions results in less variety of ADL response. Although there is a reduction in ADL requisite variety, there is a need for ADLs to be adaptive to the patient's underlying state of stability. For example, if a high sodium day results in elevated filling pressures but not yet decompensation, then the patient should be able to recognize this and have a lower sodium day after it to help compensate. Behavior assists with stability. Patients with chronic HF must reorganize their lives—how they meet their needs based on capacity and based on their underlying reduction in physiological stability—and their behavior needs to change and be more adaptive. Most research has addressed the need to change, and some has assessed factors related to adherence to behaviors intended to increase stability and lower the likelihood of instability (decompensation). Patients presenting in the ED with an HF exacerbation tended to have a self-care adherence score of 5.1 (with 10 reflecting best adherence) and knowledge about HF significantly below the cut point for adequate knowledge about the disease condition.[29] There was a positive association between the belief that HF behavior is a lifelong requirement and self-care adherence.[29] However, because the sample in this study was cross-sectional and patients were presenting in the ED for HF exacerbation, the study has no temporal sequence or comparison group to test the causal factors underlying these associations between disease knowledge and beliefs with self-care adherence and exacerbation.

It is quite plausible that not only knowledge but also belief in need and effectiveness are required for self-care adherence. In addition to knowledge and belief, patients should be capable. Being physically capable is most clearly assessed by the ability to perform ADLs to meet self-care needs. A recent study has demonstrated in a large sample of heterogeneous patients with chronic HF that knowledge was not the only determinant of self-care in patients with psychological distress.[30] Patients with high knowledge and poor self-care had worse NYHA functional class, greater depression and anxiety, and lower levels of perceived control. In the multivariate analysis, knowledge, depression symptoms, and perceived control were statistically significant predictors of self-care adherence. Similarly, Chen et al[31] demonstrated that in patients with chronic HF, health literacy and HF knowledge were associated. However, neither was associated with self-care adherence. Self-efficacy had the strongest association with self-care adherence. Interestingly, both psychological constructs of self-efficacy and perceived control are related to stress response, appraisal, and coping.[32]

Stress response and adaptability is enhanced through social support.[33,34] In an integrative review of 13 studies on the impact of social support on self-care behaviors in patients with chronic HF, Graven and Grant[35] demonstrated that 4 types of social support positively influenced self-care behaviors. Emotional support, instrumental support, informational support, and appraisal support all interacted to promote adherence. If we consider this range of support, we see that they fulfill various requirements for doing ADLs to meet needs. Emotional support is needed to keep someone encouraged, instrumental support to provide assistance when needed so there are no unmet needs, informational support to help when the patient does not know which course of action to take, and finally appraisal support to help the patient assess his or her situation to identify an unmet meet possibly requiring action. In other words, social support improves the requisite variety of ADLs. It is important to keep in mind that results thus far on self-care–only interventions have not been effective at reducing hospitalization in patients with HF.[16,36] At the same time, the interventions provided were not attempting to change self-efficacy, perceived control, or quality of life before self-care, and self-care adherence was not measured. Trials are warranted to test whether the most effective causal pathway is from self-efficacy, perceived control, quality of life, and added social support to adherence of self-care. In addition, the role of exercise to increase actual physical capacity to enable physical work to be performed and leisure activities to be participated in should be considered as strong interactive components.

Participation. The vision statement of the American Physical Therapy Association defines the need for therapists to transform the society by optimizing movement to improve the human experience.[5] Being physically capable to move and actually moving are quite different things. In patients with HF, we usually start with the first challenge, increasing their capability to move. The second challenge is for patients to actually move. Borland et al[37] demonstrated that after a 3-month group intervention, despite increases in physical capacity (increased 6-minute walk distance)

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and quality of life, participants did not significantly increase their daily physical activity. Improving stability at the participation level is about interventions that address daily participation and engagement to improve the human experience. It is possible that the subjects in the Borland et al[37] study were already functioning at the level they would want to maintain. These authors studied a group of 53 patients in NYHA functional class II and III with relatively stable chronic HF. Only 6 dropped out, 3 due to medical issues in 3 months indicating that they were, at the start, a relatively stable group of patients with chronic HF. Patients may not have increased their physical capacity to move to a level to warrant a jump in participation. For example, subjects had an average increase in the 6-minute walk test of 56 m.[37] But the level of physical function gained by most subjects may not have been enough to correspond to participation in activities such as cross-country skiing (note that the study was conducted in Sweden). Interventions to improve exercise capacity should be followed with interventions to help patients find activities that they can now do that they previously could not do and activities that they enjoy doing. Gomez-Neto et al[38] conducted a systematic review of dance therapy in patients with chronic HF. Although only 2 studies met their inclusion criteria, they demonstrated a significant increase in maximal Vo_2 and health-related quality of life. Most remarkable was the pooled and homogenous effect size for quality of life (effect size [95% CI] = 2.09 [1.65–2.54]).[38] These subjects were happy after their dance therapy! Compared with the recent Cochrane review of exercise-based therapy in patients with chronic HF on quality of life (pooled effect size = 0.56 [0.30–0.82]), the subjects with dance therapy were downright thrilled with life (their quality of life change was almost 4 times higher than the subjects in exercise).[39] Unfortunately, the studies in the Gomez-Neto et al[38] review did not conduct a long-term follow-up or measure pre- to post-physical activity as Borland et al[37] had done. However, it is not too far of a leap to expect that people who could not dance, but who enjoy dancing, now can dance and will find a way to continue dancing.

Quality of life and self-efficacy are associated with one another.[40] With such high-level constructs, the causal association could reasonably be in either direction and the directions are not mutually exclusive. It is reasonable to expect that they support one another—higher quality of life supports higher self-efficacy and higher self-efficacy supports higher quality of life. Assuming that is the case, it is possible that an intervention that improves quality of life improves self-efficacy, particularly if the intervention includes other features that could improve self-efficacy such as knowledge, health literacy, perceived control, and demonstrated success. Such an intervention could be highly valuable for reducing readmission rates in patients with HF. We do know that intervention to improve self-care adherence alone does not result in an improvement in quality of life, at least not in a relative stable sample (n = 902) of patients with chronic HF.[41] Of course, a mediating factor not assessed in this study was whether self-care adherence actually improved or if self-care did improve whether it resulted in a change in disease status or living situation.

Barriers to self-care among elderly patients with chronic HF have also been assessed using an ergonomic "Patient Work System" approach.[42] Barriers were identified across 4 components: person, task, tool, and context. Person-related barriers were related to underlying physical conditions, knowledge deficits, preferences, and perceptions. Task barriers include difficulty, complexity, ambiguity, conflicts, and consequences. Tool barriers were related to access, usability, and impact. Contextual barriers were found to be in 3 domains (physical–spatial, social–cultural, and organizational) and multiple spaces (home, community, clinic, work/school, friend). Holden et al[42] provide a useful framework for anyone considering the design of an intervention program for practice or study. It is a useful reminder that the factors influencing adherence to behavior are never simply that a patient refuses to listen.

Wingham et al[43] conducted a meta-ethnographic review of qualitative studies investigating attitudes, beliefs, expectations, and experiences of self-management strategies in people with chronic HF. From this work, they identify reaction phases that these patients seem to pass through on their way toward a particular response pattern: disruption, sense making, reaction, response, and assimilation. These reaction phases generally result in people becoming a strategic avoider, a selective denier, a well-intentioned manager, or an advanced self-manager. This article, combined with the one by Holden et al,[42] is highly recommended for anyone interested in considering development and testing of interventions to improve adherence. Factors related to and influencing whether someone ends up being a strategic avoider versus an advanced self-manager or factors that help people change from a pattern that has less effective requisite variety (strategic avoider or selective denier) to a pattern more effective (advanced self-manager) could be very valuable to explore.

SUMMARY

As this review attempts to demonstrate, 2014 was a year that emphasized a greater understanding of readmissions in patients with HF through the development of a variety of prediction models and testing of approaches to identify decompensation and improve stability. Physical therapists have much to learn from an understanding of these models to recognize their role in implementing targeted interventions to reduce readmissions. Analysis of the models reveals a lack of consideration of unmet ADLs, function, and self-care as specific predictors of readmission. The paucity of evidence in this area provides physical therapists with a unique opportunity to investigate the role of function and self-care in predicting readmission risk in patients with HF.

We have operationally defined the concepts of stability and decompensation that lead to HF exacerbations and

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hospital readmissions. The alterations in body structure and function caused by HF in multiple organs including the heart, muscle, and lungs cumulatively reduce requisite variety, thereby challenging physiological stability. Loss of control and physiological stability further limit an individual's ability for activity and participation. Physical therapists have a unique opportunity to teach patients an approach of compensating for their control limitations by using behavior-changing strategies to increase high-level requisite variety.

The ICF model specifically calls for the investigation of environmental barriers and facilitators that affect overall function. Current models fail to provide insight into the role of postdischarge environment and its effect on readmissions. Physical therapists are well positioned to address the role of environment and person–environment interactions on HF readmissions.

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